

Companion LXXXIV: Joint Constraints on α Using Multi-Scale Topological Transport and Minkowski Functionals in the Relaxation-Driven Cyclic Cosmology

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Abstract

Topological transport statistics and Minkowski functionals provide complementary probes of the large-scale structure encoded in weak-lensing convergence fields. Building on the transport-based parameter inference developed in previous Companions, this work constructs joint constraints on the relaxation parameter α in the Relaxation-Driven Cyclic Cosmology (RDCC) using a combined topology–morphology likelihood. Transport distances between persistence-diagram ensembles capture non-Gaussian topological deformations, while Minkowski functionals quantify morphological changes in the convergence field. We derive a joint Fisher matrix that incorporates cross-covariances between these statistics and identify the angular scales and topological features that contribute most strongly to constraining α . This Companion provides the first unified multi-probe framework for RDCC parameter inference from Euclid-like surveys.

1 Introduction

Weak-lensing convergence maps contain rich non-Gaussian information about the large-scale structure of the Universe. Persistence diagrams capture the topology of these maps, while Minkowski functionals quantify their morphology. Previous Companions developed a hierarchy of transport-based topological statistics and constructed Fisher forecasts for the RDCC relaxation parameter α .

The present Companion unifies these approaches by combining topological transport statistics with Minkowski functionals in a joint likelihood. This multi-probe framework leverages the complementary sensitivity of topology and morphology to RDCC-induced deformations, yielding significantly improved constraints on α .

2 Topological Transport Statistics

Let $\mathcal{T}_\ell(\alpha)$ denote the scale-resolved transport discrepancy between Λ CDM and RDCC persistence diagrams, as defined in Companion LXXXII. These statistics capture:

- shifts in birth–death structure,
- deformation of loop and void persistence,
- non-Gaussian topological signatures of RDCC.

Their sensitivity to α is encoded in the derivative $\partial\mathcal{T}_\ell/\partial\alpha$.

3 Minkowski Functionals as Morphological Probes

For a convergence field $\kappa(\theta)$, the three Minkowski functionals in 2D are:

$$V_0(\nu) = \text{area fraction above threshold}, \quad (1)$$

$$V_1(\nu) = \text{boundary length}, \quad (2)$$

$$V_2(\nu) = \text{Euler characteristic}. \quad (3)$$

RDCC modifies these functionals through the scale-dependent suppression of the gravitational potential. The derivatives $\partial V_i(\nu)/\partial\alpha$ quantify their sensitivity to α .

4 Joint Likelihood and Fisher Matrix

We define the combined data vector

$$\mathbf{d}(\alpha) = \{\mathcal{T}_\ell(\alpha), V_0(\nu), V_1(\nu), V_2(\nu)\}.$$

The joint Fisher matrix is

$$F_{\alpha\alpha} = \left(\frac{\partial\mathbf{d}}{\partial\alpha}\right)^T \mathbf{C}^{-1} \left(\frac{\partial\mathbf{d}}{\partial\alpha}\right), \quad (4)$$

where $\mathbf{C}$ includes:

- auto-covariances of transport statistics,
- auto-covariances of Minkowski functionals,
- cross-covariances between topology and morphology.

The forecasted constraint is $\sigma(\alpha) = F_{\alpha\alpha}^{-1/2}$.

5 Complementarity of Topology and Morphology

Transport statistics and Minkowski functionals probe different aspects of the convergence field:

- Transport statistics are sensitive to the *connectivity structure* of peaks, filaments, and voids.
- Minkowski functionals probe *shape, area, and curvature*.

RDCC induces:

- topological shifts in loop and void persistence,
- morphological changes in peak abundance and curvature,
- scale-dependent deformations that appear differently in both probes.

The joint analysis captures these complementary signatures.

6 Conclusion

Topological transport statistics and Minkowski functionals provide complementary probes of RDCC-induced deformations in weak-lensing convergence fields. By combining them in a joint Fisher analysis, we obtain significantly improved constraints on the relaxation parameter α . This multi-probe framework bridges the gap between geometric transport theory and observational cosmology, and is directly applicable to Euclid-like surveys.

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